Chapter 7: Questions, Answers & Rationales

Comprehensive study bank covering General Pathology, Genetics, and Clinical Case Studies.

Case A

Mr. Steve Perry is a new patient to your dental practice. He presents with generalized gingivitis indicated by gingival inflammation, edema, bleeding upon probing, and red fiery tissue. You also notice generalized heavy biofilm at the gingival margin. As you begin to explain your clinical findings to Mr. Perry, he explains he often feels discomfort and pain when flossing his teeth so he has discontinued his flossing routine. Use Case A to answer questions 1 to 4.

Question 1

What is the cause of the redness described in Case A?

a. Vasoconstriction of blood vessels
b. Increased vascularity
c. Elimination of malignant neoplasia as a possible cause
d. Exudation of fluid

Correct Answer: B. Increased vascularity is the cause of redness

Clinical Rationale:
Increased vascularity is the cause of redness.

* a. Vasoconstriction of blood vessels causes stasis.
* c. Stretching of pain receptors causes pain.
* d. Exudation of fluid causes swelling.

Question 2

What is the cause of the gingival edema described in Case A?

a. Vasoconstriction of blood vessels
b. Increased vascularity
c. Stretching of pain receptors
d. Exudation of fluid

Correct Answer: D. Exudation of fluid causes gingival edema

Clinical Rationale:
Exudation of fluid causes gingival edema.

* a. Vasoconstriction of blood vessels causes stasis.

* b. Increased vascularity causes redness.
* c. Stretching of pain receptors causes pain.

Question 3

Which of the following signs of inflammation can explain the pain Mr. Perry is reporting?

a. Rubor

b. Calor

c. Dolor

d. Functio laesa

Correct Answer: C. Dolor is responsible for pain

Clinical Rationale:
Dolor is responsible for pain.

* a. Rubor is redness.
* b. Calor is heat.
* d. Functio laesa is loss of function.

Question 4

Mr. Perry's gingival tissue is reacting in response to an inflammatory stimulus. The inflammatory response has what type of reaction?

a. Vascular and cellular reaction

b. Vascular and regenerative reaction

c. Regenerative and reparative reaction

d. Cellular and regenerative reaction

Correct Answer: A. Mr. Perry’s gingival tissue is reacting to a vascular and cellular reaction in the inflammatory process

Clinical Rationale:
The inflammatory response primarily manifests as a vascular and cellular reaction to stimulus.

* b. Regenerative reaction is not part of the inflammatory response.
* c. Regenerative and reparative reactions are part of the wound healing process, not the inflammatory response.
* d. Regenerative reaction is not part of the inflammatory response.

Case B

Mrs. Evelyn Goddard presents with a periodontal abscess on the buccal mucosa adjacent to tooth #30. She noticed swelling and mild pain in this area when she woke up in the morning and called the dental office immediately for an appointment. Upon examination with the dental hygienist, the area was significantly enlarged with redness and edema. A fistula was present. When the tissue was compressed, suppuration was expressed. Use Case B to answer questions 5 to 10.

Question 5

The body's response to the development of an abscess most likely represents:

a. Acute inflammation

b. Chronic inflammation

c. Regeneration

d. Repair

Correct Answer: A. Acute inflammation manifests with exudation of fluid (edema in this case) and emigration of leukocytes, mainly neutrophils, and is characterized by rapid onset and short duration

Clinical Rationale:
Acute inflammation is characterized by rapid onset, short duration, exudation of fluid (edema), and neutrophil emigration.

* b. Chronic inflammation is a sustained response over a longer duration.
* c. Regeneration represents growth of cells and tissues to replace lost structures.
* d. Repair is a combination of regeneration and scar formation.

Question 6

Which of the following is another name for redness (as seen in Mrs. Goddard's case)?

a. Dolor

b. Calor

c. Tumor

d. Rubor

Correct Answer: D. Rubor is the other name for redness

Clinical Rationale:
Rubor represents redness.

* a. Dolor is pain.
* b. Calor is heat.
* c. Tumor is swelling.

Question 7

Edema is associated with which of the following processes?

a. Redness caused by increased vascularity

b. Exudation of fluid

c. Dilation of blood vessels

d. Stretching of pain receptors

Correct Answer: B. Edema is associated with exudation of fluid also known as increased fluid in the interstitial space

Clinical Rationale:
Edema is caused by the exudation of fluid into the interstitial tissue spaces.

* a. Redness is caused by increased vascularity.
* c. Calor (heat) is associated with blood flow and chemical mediators.
* d. Dolor (pain) is caused by stretching of pain receptors.

Question 8

Which of the following cells are primary white blood cells associated with suppuration (pus formation)?

a. Lymphocytes and plasma cells

b. Eosinophils and mast cells

c. Neutrophils and macrophages

d. Eosinophils and lymphocytes

Correct Answer: C. Neutrophils and macrophages are the primary white blood cells associated with suppuration; neutrophils are the first white blood cells to emigrate to the site of injury and the primary cell involved in acute inflammation followed by macrophages

Clinical Rationale:
Neutrophils and macrophages are the cells that form pus. Neutrophils arrive first during acute inflammation, followed by macrophages.

* a. Lymphocytes and plasma cells are involved in chronic inflammation.
* b. Eosinophils are involved in immune reactions/parasitic infections; mast cells are involved in allergic responses.

Question 9

Recent evidence demonstrates that in addition to phagocytosis, neutrophils also perform which function?

a. Regulate vascular permeability

b. Release chemical mediators

c. Regulate bronchial smooth muscle tone

d. Contribute to the repair process

Correct Answer: D. Neutrophils contribute to repair as well as the inflammatory response

Clinical Rationale:
Neutrophils perform phagocytosis and also actively contribute to the wound repair process.

* a, b, and c are all primary functions of mast cells, not neutrophils.

Question 10

What type of healing will occur in the narrow space left by the fistula/abscess drainage?

a. Primary intention

b. Secondary intention

c. Tertiary intention

d. None of the above

Correct Answer: A. Primary intention healing will occur. Even though sutures are not placed, the edges of the wound will be in close approximation and there will be a narrow incisional space for healing to occur

Clinical Rationale:
Primary intention occurs when wound edges are close together, leaving a narrow space for healing.

* b. Secondary intention involves a large, separated space with abundant granulation tissue.
* c. and d. represent tertiary intention which is delayed wound closure due to contamination.

Case C

Sam Adams presents for extraction of tooth #32 which is impacted. The extraction is performed and surgery is uneventful. Sutures are placed and the patient is discharged with instructions to alternate acetaminophen and ibuprofen as needed for pain, apply ice on the outside of the mouth to reduce swelling, gently rinse the mouth with a mild antiseptic mouth rinse, consume liquid based foods/soft diet for 1-2 days and return in one week for post-op evaluation and suture removal. Use Case C to answer questions 11 to 17.

Question 11

What is the first step in the healing process of the extraction socket?

a. An inflammatory process

b. Thrombus or clot formation

c. Granulation tissue formation

d. Collagen remodeling

Correct Answer: B. Thrombus or clot is formed at the site of injury as the first step in the healing process

Clinical Rationale:
Clot/thrombus formation is always the very first step in wound healing.

* a. Inflammation occurs shortly after clot formation.
* c and d are later stages in the wound healing process.

Question 12

During wound healing, the epithelium will form a basement membrane under the scab within what time frame?

a. 12 hours

b. 24 hours

c. 24-48 hours

d. 72 hours

Correct Answer: C. Epithelium will form a basement membrane under the scab within 24-48 hours

Clinical Rationale:
Epithelial migration and basement membrane formation under the protective scab occur within 24 to 48 hours post-injury.

Question 13

By day 5 in the wound healing process, granulation tissue fills the incisional space. All of the following are characteristics of granulation tissue EXCEPT one. Which one is the EXCEPTION?

a. Tissue is pink and soft

b. New small blood vessels appear

c. New vessels are fully patent and do not leak

d. Amount of granulation tissue depends on the size of the wound

Correct Answer: C. New vessels tend to leak and are edematous

Clinical Rationale:
During early granulation, newly formed blood vessels are immature, fragile, and leak fluid, making the tissue edematous. They are NOT fully patent and leak-free.

* All other options (a, b, and d) are correct characteristics of granulation tissue.

Question 14

Tissues recover what percentage of tensile strength over a 3-month period of healing as compared with intact skin?

a. 60-70%

b. 70-80%

c. 80-90%

d. 100%

Correct Answer: B. Tissues recover 70-80% of tensile strength over a 3-month period of healing as compared with intact skin

Clinical Rationale:
Healed wounds recover up to 70-80% of original tensile strength by 3 months. Strength rarely reaches 100% of uninjured tissue.

Question 15

Which of the following is considered a systemic factor (rather than a local factor) that may inhibit or prolong the wound healing process?

a. Size of the wound
b. Foreign material in the wound
c. Location of the wound
d. Blood supply

Correct Answer: D. Blood supply is considered a systemic factor that may inhibit or prolong the wound healing process

Clinical Rationale:
Blood supply (vascular status) is a systemic parameter affecting healing capacity.

* a, b, and c are local factors that directly affect only the localized wound site.

Question 16

Which wound healing anomaly occurs as a result of deficient scar formation?

a. Wound dehiscence
b. Keloid
c. Contracture
d. Fibrosis

Correct Answer: A. Wound dehiscence occurs with deficient scar formation

Clinical Rationale:
Wound dehiscence (re-opening of a wound) occurs when scar tissue formation is deficient.

* b. Keloid is an overgrowth of scar tissue.
* c. Contracture is an exaggeration of normal wound contraction.
* d. Fibrosis is extensive deposition of collagen that causes dysfunction.

Question 17

Local factors that could prolong the wound healing of the extraction site include all of the following EXCEPT one. Which one is the EXCEPTION?

a. Size of the wound

b. Blood supply

c. Foreign material in the wound

d. Infection of the wound

Correct Answer: B. Size, location and type of wound is a local factor for wound healing

Clinical Rationale:
Blood supply is a systemic factor that affects wound healing, whereas size, location, foreign material, and infection are local factors.

Case D

Mrs. Monica Salt is a female patient scheduled for periodontal maintenance. As you are reviewing her health history, Mrs. Salt tells you that she is pregnant. She is concerned about her baby being born with a cleft lip and palate because there is a family history. Use Case D to answer questions 18 to 21.

Question 18

Which of the following is a primary indication for genetic counseling?

a. Routine pregnancy checkup

b. Advanced maternal age only

c. Maternal nutrition concern

d. Previous family history of cleft lip and palate

Correct Answer: D. Previous family history of cleft lip and palate would be reasons for genetic counseling

Clinical Rationale:
A family history of congenital disorders like cleft lip and palate is a primary indication for genetic counseling.

* a, b, and c do not warrant chromosome analysis or genetics referrals.

Question 19

The majority of mutations associated with hereditary diseases are:

a. Genome mutations

b. Chromosome mutations

c. Submicroscopic gene mutations

d. Cytosine gene mutations

Correct Answer: C. A majority of mutations associated with hereditary diseases are submicroscopic gene mutations

Clinical Rationale:
Submicroscopic gene mutations (point mutations or small indels) make up the majority of genetic alterations causing hereditary disease.

Question 20

In this case, genetic testing for the fetus can be performed on cells obtained by:

a. Amniocentesis

b. Chorionic villus biopsy

c. Umbilical cord blood

d. All of the above

Correct Answer: D. Amniocentesis, chorionic villus biopsy, and umbilical cord blood are ways in which genetic testing can be performed on cells obtained from a fetus

Clinical Rationale:
All listed methods (amniocentesis, CVS, and cordocentesis) obtain fetal cells for karyotyping or genetic analysis.

Question 21

Cleft lip and/or cleft palate is considered a disorder of:

a. Single gene inheritance

b. Multiple factorial inheritance

c. Autosomal dominant inheritance

d. Autosomal recessive inheritance

Correct Answer: B. Cleft lip and/or cleft palate is considered a disorder of multiple factorial inheritance

Clinical Rationale:
Cleft lip and palate results from multiple genetic and environmental factors (multifactorial inheritance).

* a, c, and d represent single-gene inheritance patterns (such as Huntington's or sickle cell anemia).

Case E

Mr. and Mrs. Dorchester present to the dental practice for a consult. They would like to bring their 4-year-old son to the office for comprehensive oral health care; however, he has Duchenne muscular dystrophy. They are concerned that the practice can provide care managing their son’s condition as the disease progresses. Use Case E to answer questions 22 to 25.

Question 22

What type of genetic disease does Duchenne muscular dystrophy represent?

a. Autosomal dominant disorder

b. Autosomal recessive disorder

c. X-linked recessive disorder

d. Disorder with multifactorial inheritance

Correct Answer: C. Duchenne muscular dystrophy is an X-linked recessive disorder

Clinical Rationale:
Duchenne muscular dystrophy is inherited in an X-linked recessive pattern.

* a. Neurofibromatosis is autosomal dominant.
* b. Phenylketonuria is autosomal recessive.
* d. Coronary heart disease is multifactorial.

Question 23

All of the following are characteristics of Duchenne muscular dystrophy EXCEPT one. Which one is the EXCEPTION?

a. Primarily affects males

b. All sons of carrier mothers are carriers

c. No male-to-male transmission

d. Affected males transmit the genes to all daughters

Correct Answer: B. All daughters are carriers of Duchenne muscular dystrophy

Clinical Rationale:
Sons of carrier mothers either inherit the mutated X chromosome and are affected, or inherit the healthy X and are unaffected; they are never carriers. All daughters of affected males will be carriers (inheriting his only mutated X chromosome).

Question 24

Which of the following is an example of a multifactorial inheritance disorder?

a. Hemophilia

b. Duchenne muscular dystrophy

c. Red-green color blindness

d. Coronary heart disease

Correct Answer: D. Coronary heart disease is an example of multifactorial inheritance disorders

Clinical Rationale:
Coronary heart disease is multifactorial, depending on multiple genes and lifestyle elements.

* a, b, and c are all X-linked recessive disorders.

Question 25

All of the following are methods of examining genetic material for Duchenne muscular dystrophy EXCEPT one. Which one is the EXCEPTION?

a. Karyotyping
b. Chromosome analysis
c. Polymerase chain reaction
d. Electromyography

Correct Answer: D. Karyotyping, chromosome analysis, and polymerase chain reaction are all methods of examining genetic material for Duchenne muscular dystrophy

Clinical Rationale:
Karyotyping, chromosome analysis, and PCR directly analyze genetic structures or DNA. Electromyography measures muscle electrical activity, not genetic material.

Case F

Janis Johnson presents to your practice with a chief complaint of a mass on the left buccal mucosa of normal tissue coloration. The lesion has been present for 2 months, is painless, but interferes occasionally with chewing activities. The patient reports that she has been known to bite the mass causing it to become ulcerated. Answer the following items as they relate to the diagnostic process. Use Case F to answer questions 26 to 30.

Question 26

Classify the abnormality based on the primary manifestation.

a. A white mucosal discoloration without loss of mucosal integrity
b. A dark mucosal discoloration
c. A loss of mucosal integrity
d. Enlargement of soft tissues

Correct Answer: D. Enlargement of soft tissues is the primary manifestation because the lesion is a mass on the buccal mucosa

Clinical Rationale:
The lesion is a localized swelling or mass, classifying it as an enlargement of soft tissues. It is not white, dark, and is not primarily an ulceration (loss of integrity) though it can be traumatized.

Question 27

Which of the following is NOT an appropriate clinical diagnostic technique to evaluate this mass?

a. Probing
b. Palpation

c. Aspiration

d. Visual examination

Correct Answer: A. A mass cannot be probed

Clinical Rationale:
A soft tissue mass is assessed by palpation, inspection, or aspiration. Probing is used to measure pocket depths or tissue defects, not solid masses.

Question 28

Alcohol use, tobacco use, and oral behaviors describe which component of the patient's history during the diagnostic process?

a. Demographics

b. Habits

c. Recent history

d. Patient awareness of condition

Correct Answer: B. Alcohol use, tobacco use, and oral behaviors describe a patient’s habits

Clinical Rationale:
These elements are grouped under patient habits.

* a. Demographics include age, gender, race.
* c. Recent history includes trauma, surgeries, infections.

Question 29

Pain, discomfort, altered function, duration, and response to factors such as stress or certain foods all describe the patient's:

a. Demographics

b. Habits

c. Recent history

d. Awareness of condition

Correct Answer: D. Pain, discomfort or altered function, duration, response to factors such as stress or certain foods all describe patient awareness of condition

Clinical Rationale:
These items capture the patient's subjective awareness of their condition.

Question 30

All of the following are terms for a working diagnosis EXCEPT one. Which one is the EXCEPTION?

a. Differential diagnosis

b. Probable diagnosis

c. Preliminary diagnosis

d. Tentative diagnosis

Correct Answer: B. Probable diagnosis is a fabricated term

Clinical Rationale:
Probable diagnosis is not a formal diagnostic term. Differential, preliminary, and tentative are standard classifications of a working diagnosis.

Question 31

Which white blood cells are responsible for producing vasoconstrictive chemical mediators that maintain chronic inflammation and performing phagocytosis?

a. Eosinophils

b. Lymphocytes

c. Macrophages

d. Mast cells

Correct Answer: C. Macrophages are the white blood cells responsible for producing vasoconstrictive chemical mediators which maintain chronic inflammation

Clinical Rationale:
Macrophages play a key role in chronic inflammation by phagocytizing tissue debris and releasing chemical mediators that sustain the inflammatory state.

* a. Eosinophils assist in allergic/parasitic responses.
* b. Lymphocytes orchestrate chronic immune responses.
* d. Mast cells trigger acute inflammation via histamine release.

Question 32

Which white blood cells are responsible for assisting in immune reactions and defending against parasitic infections?

a. Eosinophils

b. Lymphocytes

c. Macrophages

d. Mast cells

Correct Answer: A. Eosinophils are involved in immune reactions and parasitic infections

Clinical Rationale:
Eosinophils are specialized granulocytes key to fighting parasitic infections and regulating immediate allergic

hypersensitivity.

Question 33

Which chemical mediator is responsible for causing dilation of arterioles, constricting large arteries, increasing vascular permeability (edema), and smooth muscle contraction?

a. Serotonin
b. Bradykinin
c. Histamine
d. Arachidonic acid

Correct Answer: C. Histamine is the chemical mediator that is responsible for causing dilation of arterioles and constricting large arteries, edema and smooth muscle contraction

Clinical Rationale:
Histamine (predominantly from mast cells) dilates arterioles, constricts larger arteries, increases vascular permeability, and contracts smooth muscle.

* a. Serotonin causes vasoconstriction/increased permeability in immune responses.
* b. Bradykinin dilates vessels, increases permeability, and causes pain.
* d. Arachidonic acid is a precursor to prostaglandins/leukotrienes.

Question 34

Programmed cell death that occurs during phagocytosis without inducing inflammation is referred to as:

a. Engulfment
b. Degradation
c. Apoptosis
d. Chemotaxis

Correct Answer: C. Apoptosis is programmed cell death in phagocytosis

Clinical Rationale:
Apoptosis is programmed, non-inflammatory cell death.

* a, b, and d are steps in the migration and ingestion stages of phagocytosis.

Question 35

Pro-resolving mediators (like lipoxins and resolvins) perform all of the following functions EXCEPT one. Which one is the EXCEPTION?

a. Evoke potent antiinflammatory mechanisms
b. Return cells to homeostasis

c. Enhance microbial clearance

d. Induce pain

Correct Answer: D. Pro-resolving mediators help to reduce pain

Clinical Rationale:
Pro-resolving mediators function to resolve inflammation, clear microbes, and *reduce* pain. They do not induce pain.

Question 36

Which of the following chemical mediators moderates the susceptibility, development, and progression of autoimmune and inflammatory diseases?

a. Interleukin-6 (IL-6)

b. Tumor Necrosis Factor (TNF)

c. Interleukin-1 (IL-1)

d. Nitric Oxide (NO)

Correct Answer: A. IL-6 is the chemical mediators moderates the susceptibility, development and progression of autoimmune and inflammatory diseases

Clinical Rationale:
IL-6 is a pleiotropic cytokine that regulates chronic inflammation and autoimmune progression.

* b and c (TNF and IL-1) drive acute systemic reactions and septic shock.
* d (NO) is microbicidal and vasoactive.

Question 37

All of the following are possible outcomes of acute inflammation EXCEPT one. Which one is the EXCEPTION?

a. Complete resolution

b. Healing by connective tissue replacement (fibrosis/scarring)

c. Progression of tissue to chronic inflammation

d. Leukopenia

Correct Answer: D. Leukopenia is not a possible outcome of inflammation; leukocytosis is the outcome of inflammation

Clinical Rationale:
Acute inflammation leads to leukocytosis (elevated WBCs), not leukopenia (depleted WBCs). Complete resolution, scarring, and chronic progression are normal potential outcomes.

Question 38

Which of the following oral conditions is associated with chronic inflammation?

a. Gingivitis

b. von Willebrand disease

c. Anemia

d. Caries

Correct Answer: A. Gingivitis is associated with chronic inflammation

Clinical Rationale:
Gingivitis and periodontitis represent chronic inflammatory responses to plaque biofilm.

* b is a genetic bleeding disorder.
* c is a blood disorder.
* d is a localized infectious decay process.

Question 39

A hallmark of healing and repair is the formation of:

a. A scab

b. Granulation tissue

c. Adjacent blood vessels

d. Endothelial growth factor

Correct Answer: B. Formation of granulation tissue is a hallmark of healing

Clinical Rationale:
Granulation tissue, rich in new capillaries and fibroblasts, is the diagnostic hallmark of active healing.

Question 40

Extensive deposition of collagen that sustains the synthesis and secretion of growth factors during the wound healing process causing significant scar formation represents:

a. Wound dehiscence

b. Keloid

c. Contracture

d. Fibrosis

Correct Answer: D. Fibrosis is extensive deposition of collagen that sustains the synthesis and secretion of growth factors during the wound healing process causing significant scar formation

Clinical Rationale:
Fibrosis refers to extensive, pathologic deposition of collagen leading to dense scar tissue.

* a is split-open margins.
* b is a raised, overgrown scar.
* c is pathologic shrinkage.

Question 41

Poor wound healing associated with a severely compromised blood supply can lead to:

a. Chronic obstructive pulmonary disease (COPD)

b. Rheumatoid arthritis

c. Amputation

d. Cirrhosis

Correct Answer: C. Poor wound healing with a compromised blood supply can lead to amputation

Clinical Rationale:
Severe ischemia (lack of blood flow) prevents healing, leading to tissue necrosis/gangrene and eventual amputation.

Question 42

Stem cells derived from another individual of the same species are referred to as:

a. Autogenous

b. Allogenous

c. Unipotent

d. Totipotent

Correct Answer: B. Stem cells derived from other individuals is defined as allogenous

Clinical Rationale:
Allogenous cells come from a separate donor of the same species. Autogenous cells come from the patient's own body.

Question 43

Characteristics of embryonic stem cells include all of the following EXCEPT one. Which one is the EXCEPTION?

a. Pluripotent capacity

b. Isolated from the inner cell mass of blastocysts

c. Found in adult skin, the lining of the intestine, cornea, and brain

d. May be used to treat organs affected by diabetes or spinal cord injury

Correct Answer: C. Adult stems are found in skin, lining of intestine, cornea and brain

Clinical Rationale:
Stem cells in the skin, gut, cornea, and brain are multipotent adult stem cells, not embryonic stem cells. The other properties describe embryonic stem cells.

Question 44

What is the most common commercially available source of adult stem cells?

a. Umbilical cord blood stem cells

b. Dental stem cells

c. Adipose-derived stem cells

d. Bone marrow-derived stem cells

Correct Answer: D. Bone marrow-derived stem cells are the most common commercially available stem cell

Clinical Rationale:
Bone marrow aspirates are the most widely used and commercially available source of adult hematopoietic and mesenchymal stem cells.

Question 45

A common, unique property of stem cells is their:

a. Ability to undergo rapid senescence

b. Capacity to intermittently undergo division or remain quiescent

c. Development from terminally differentiated cells

d. Production of single-lineage cells only

Correct Answer: B. A common property of stem cells is the capacity to intermittently undergo division or remain quiescent

Clinical Rationale:
Stem cells can replicate indefinitely (self-renewal) or enter a resting state (quiescence) until triggered to divide.

Question 46

Sources of dental stem cells include all of the following EXCEPT one. Which one is the EXCEPTION?

a. Pulp chamber walls (dentin)

b. Periodontal ligament

c. Unerupted third molars

d. Exfoliated primary teeth

Correct Answer: A. The pulp chamber is not a source of dental stem cells

Clinical Rationale:
Dental stem cells are harvested from soft tissues like the dental pulp, PDL, follicle, or apical papilla. The mineralized dentin walls of the pulp chamber do not contain stem cells.

Question 47

Approximately how many genes do humans have?

a. 10,000 to 15,000
b. 15,000 to 20,000
c. 20,000 to 25,000
d. 25,000 to 30,000

Correct Answer: C. Humans have 20,000 to 25,000 genes

Clinical Rationale:
Current sequencing estimates place the human genome at approximately 20,000 to 25,000 protein-coding genes.

Question 48

Which of the following represents the four letters of the DNA code?

a. Adenine, thiamine, cytosine, guanine
b. Adenine, thymine, cystine, guanine
c. Adenosine, thymine, cytosine, guanine
d. Adenine, thymine, cytosine, guanine

Correct Answer: D. Adenine, thymine, cytosine, and guanine are the four letters of the DNA code

Clinical Rationale:
DNA nitrogenous bases are Adenine, Thymine, Cytosine, and Guanine. Other options use incorrect terms (thiamine is a vitamin, cystine is an amino acid, adenosine is a nucleoside).

Question 49

For an autosomal dominant inheritance disorder, a person has what percent chance of passing the trait to their offspring?

a. 25%
b. 50%
c. 75%
d. 100%

Correct Answer: B. A person has a 50% chance of passing a trait to an offspring for an autosomal dominant inheritance disorder

Clinical Rationale:
A heterozygote with an autosomal dominant condition has a 50% chance of passing the mutant allele to each child.

Question 50

With autosomal recessive inheritance, two carrier parents have what percent chance with each conception of having an affected child?

a. 25%

b. 50%

c. 75%

d. 100%

Correct Answer: A. Two carrier parents have a 25% chance of having an affected child with an autosomal recessive inheritance disorder with each conception

Clinical Rationale:
For an autosomal recessive trait, each pregnancy of carrier parents (Aa x Aa) has a 25% chance of producing an affected child (aa).
